

Program : Diploma in Wood and Paper Technology	
Course Code : 3125	Course Title: Electrical Technology Lab
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To familiarize students the practical execution of fundamental theories of Electrical Engineering
- To familiarize students with the working of electrical instruments and electronic components

Course Prerequisites:

Topic	Course code	Course name	Semester
Electrical instruments, Electronic Devices.		Applied Physics-I	1
		Applied Physics-II	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate measurement of parameters using various meter	9	Applying
CO2	Illustrate characteristics of various electrical circuit	9	Applying
CO3	Analyse the performance of electrical machines	18	Applying
CO4	Identify different electronic components of electrical instruments.	6	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2				3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate measurement of parameters using various meter		
M1.01	Assemble and test a simple electric circuit containing source, switch, fuse and a load.	3	Applying
M1.02	Measure the voltage and current of a DC circuit including following steps. a. Selection of proper instrument and range b. Reading of meter in correct position and correct way. c. Use of multi range meter. (to clarify the importance of multiplication factor)	3	Applying
M1.03	Assemble following circuits and verify voltage and current divisions a. Series circuit contains three lamps of different power ratings. b. Parallel circuit contains three lamps of different power ratings	3	Applying
CO2	Illustrate characteristics of various electrical circuit		
M2.01	Measure a wattmeter connected to a given load. Measure an energy meter connected in an existing installation	3	Applying
M2.02	Measure the armature and field resistance of a DC shunt motor	3	Applying
M2.03	Measure the current of 100 W incandescent lamp when burning Verify Ohm's law	3	Applying
	Lab Exam – I	1.5	

CO3	Analyse the performance of electrical machines		
M.3.01	Start the shunt, compound and series motors using starters Conduct the speed - torque characteristics of D.C shunt motors Reverse the direction of rotation of DC motors Measure the voltage, current and power of a single phase inductive load and calculation of inductance Start the slip ring induction motor Reverse the direction of rotation of single phase and three phase induction motors	6	Applying
M.3.02	Conduct the ratio test and polarity test of single phase transformer Load the single phase transformer and determine the efficiency Calibrate the single phase energy meter Plot the magnetization characteristics of a given D.C. shunt generator Measure 3 Phase power by using two watt meter. Perform the O.C and S.C test on a single phase transformer and pre- determine the efficiency Plot the external characteristics of a D.C. shunt generator	12	Applying
CO4	To understand different electronic components of Electrical instruments.		
M.4.01	Familiarise electronic components of : UTM (Universal Testing Machine) Ammeter, Volt meters, Watt meters, Multi meters, Shunt and compound and series motors	6	Applying
	Lab Exam – II	1.5	

Text / Reference:

T/R	Book Title/Author
T1	A text book of Electrical technology, B L Theraja and A K Theraja
R1	Electrical machines, by J B Gupta
R2	EW. Golding Electrical Measurements and Measuring Instruments, 5th ed., Reem publications, 2011
R3	V.K. Mehta & Rohit Mehta, "Principles of Electrical Engineering and electronics", S. Chand Publication, 2010
R4	N.N. Bhargava , D.C. Kulshreshtha S.C. Gupta, "Basic Electronics and Linear Circuits", Tata McGraw Hill Education, 1983

Online Resources:

Sl.No	Website Link
1	www.electrical4u.com
2	https://www.vlab.co.in/ba-nptel-labs-electrical-engineering